Abel–Ruffini Theorem via an Explicit Quintic

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

A complete authored walkthrough of abel–ruffini theorem via an explicit quintic, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

MA-05 / Modern Algebra / Report v1

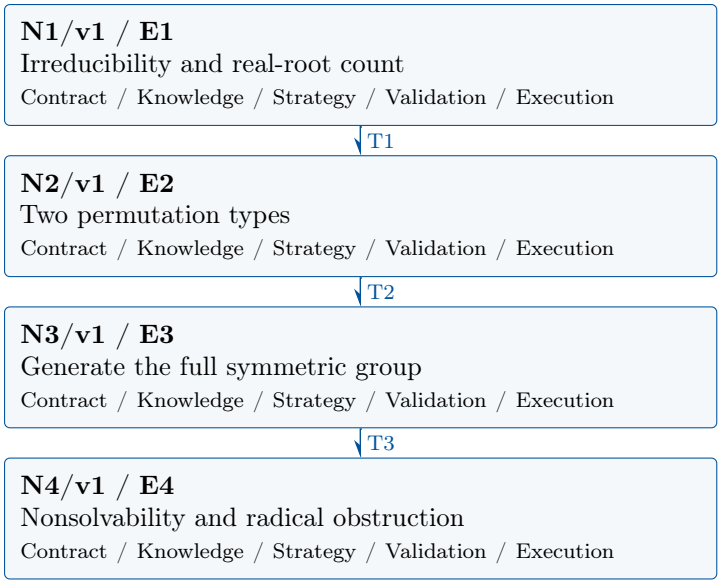
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Prove $f(x) = x^5 - 4x + 2 \in \mathbb{Q}[x]$ has Galois group S_5 and is not solvable by radicals. Deduce there is no radical formula solving every quintic.*

A. Foundations and permitted imports

Use Eisenstein's criterion: a primitive integer polynomial with all nonleading coefficients divisible by a prime ℓ, leading coefficient not divisible by ℓ, and constant coefficient not divisible by ℓ^2 is irreducible over Q. Import characteristic-zero splitting-field Galois theory, transitivity for irreducible polynomials, orbit–stabilizer, Cauchy's group theorem, Lagrange's theorem, and the radical-solvability criterion: a polynomial over Q is solvable by radicals iff its splitting-field Galois group is solvable. The permutation generation and S_5 nonsolvability steps are established below.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. An irreducible quintic with three real roots and one conjugate pair.

Receiving activity. N2: Two permutation types

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Irreducibility and real-root count; prior exports E1/v1

T2: N2/E2/v1 to N3. G contains a5-cycle and a transposition.

Receiving activity. N3: Generate the full symmetric group

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: A5-cycle σ and transposition τ in G; prior exports E2/v1

T3: N3/E3/v1 to N4. The exact Galois group is S_5.

Receiving activity. N4: Nonsolvability and radical obstruction

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: $G = S_5$; radical criterion; prior exports E3/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Irreducibility and real-root count

Object and role. Combine Eisenstein with exact critical-point signs.

Inputs. $f = x^5 - 4x + 2$; foundational imports stated in §1

C — Contract

Own E1/v1: An irreducible quintic with three real roots and one conjugate pair.

K — Knowledge

Retain $f = x^5 - 4x + 2$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Combine Eisenstein with exact critical-point signs.

V — Validation

A plotted root count alone is not an exact certificate.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Eisenstein at 2 applies: the coefficients $0, 0, 0, -4, 2$ are divisible by 2 and the constant 2 is not divisible by 4.

N1.2 Thusf is irreducible.

N1.3 Its derivative $5x^4 - 4$ vanishes at $\pm\alpha$ where $\alpha = (4/5)^{1/4}$; it is positive outside and negative inside.

N1.4 Since $\alpha > 5/8$, $f(\alpha) = 2 - (16/5)\alpha < 0$ and $f(-\alpha) = 2 + (16/5)\alpha > 0$.

N1.5 The limits at $-\infty$ and ∞ are $-\infty$ and ∞.

N1.6 Strict monotonicity on the three intervals therefore gives exactly three distinct real roots and two nonreal roots forming one conjugate pair.

N1.7 Irreducibility over Q in characteristic zero supplies separability, so all five complex roots are distinct.

Export E1/v1. An irreducible quintic with three real roots and one conjugate pair.

Acceptance gate. A plotted root count alone is not an exact certificate. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A plotted root count alone is not an exact certificate. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Two permutation types

Object and role. Read transitivity and complex conjugation.

Inputs. Irreducibility and real-root count; prior exports E1/v1

C — Contract

Own E2/v1: G contains a5-cycle and a transposition.

K — Knowledge

Retain Irreducibility and real-root count; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Read transitivity and complex conjugation.

V — Validation

A nonreal-root pair count determines the entire conjugation permutation.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Let G be the splitting-field Galois group acting faithfully on five roots.

N2.2 Irreducibility gives a transitive action, so 5 divides|G| by orbit–stabilizer.

N2.3 Cauchy supplies an element of order 5, necessarily a5-cycle in S_5.

N2.4 Complex conjugation is an automorphism of the splitting field fixing Q; it fixes the three real roots and swaps the nonreal pair, hence is a transposition.

Export E2/v1. G contains a5-cycle and a transposition.

Acceptance gate. A nonreal-root pair count determines the entire conjugation permutation. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: A nonreal-root pair count determines the entire conjugation permutation. Recheck dependent nodes N3, N4.

C. N3/v1: Generate the full symmetric group

Object and role. Use conjugate transpositions as edges in a connected graph.

Inputs. A5-cycle σ and transposition τ in G; prior exports E2/v1

C — Contract

Own E3/v1: The exact Galois group is S_5.

K — Knowledge

Retain A5-cycle σ and transposition τ in G; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Use conjugate transpositions as edges in a connected graph.

V — Validation

A cycle/transposition claim needs the connected-graph proof; it is not automatic for arbitrary cycle lengths.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Label the roots cyclically so σ acts by addition1 modulo 5, and write $\tau = (a, b)$ with nonzero differenceb $-a$ mod 5.

N3.2 Its conjugates $\sigma^j \tau \sigma^{(-j)}$ swap the translated pair $s(a + j, b + j)$.

N3.3 These edges form a connected graph since repeated addition of a nonzero residue visits all five vertices.

N3.4 Edge transpositions on a connected graph generate every transposition: along a path, conjugating the final edge swap by the preceding path swaps gives a swap between the endpoints.

N3.5 All transpositions generateS $_5$, so $S_5 \subset G \subset S_5$ and $G = S_5$.

Export E3/v1. The exact Galois group is S_5.

Acceptance gate. A cycle/transposition claim needs the connected-graph proof; it is not automatic for arbitrary cycle lengths. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: A cycle/transposition claim needs the connected-graph proof; it is not automatic for arbitrary cycle lengths. Recheck dependent nodes N4.

D. N4/v1: Nonsolvability and radical obstruction

Object and role. Prove the needed group obstruction before applying the field theorem.

Inputs. $G = S_5$; radical criterion; prior exports E3/v1

C — Contract

Own E4/v1: This explicit quintic is not solvable by radicals; no radical formula covers every quintic.

K — Knowledge

Retain $G = S_5$; radical criterion; prior exports E3/v1; keep their assumptions and versions attached.

S — Strategy

Prove the needed group obstruction before applying the field theorem.

V — Validation

Some quintics are solvable; the conclusion is not a claim about every individual quintic.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 In A_5 the conjugacy-class sizes are 1,15 (double transpositions),20 (3-cycles), and 12,12 (the two classes of 5-cycles).

N4.2 They follow from cycle counting and centralizer sizes4,3,5 in A_5.

N4.3 A normal subgroup is a union of these classes containing the identity, and its order divides60.

N4.4 The possible such sums are 1,13,25,16,28,40,21,33,45,36,48,60; only1 and 60 divide60.

N4.5 Thus A_5 is simple.

N4.6 It is nonabelian, so its nontrivial commutator subgroup, being normal, is all A_5; its derived series never reaches1.

N4.7 Hence A_5, and therefore S_5 which contains it, is not solvable.

N4.8 By the imported radical criterion the quintic is not solvable by radicals.

N4.9 A radical formula for every quintic would solve this one, contradiction.

Export E4/v1. This explicit quintic is not solvable by radicals; no radical formula covers every quintic.

Acceptance gate. Some quintics are solvable; the conclusion is not a claim about every individual quintic. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Some quintics are solvable; the conclusion is not a claim about every individual quintic. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Irreducibility and real-root count

1. Eisenstein at 2 applies: the coefficients $0, 0, 0, -4, 2$ are divisible by 2 and the constant 2 is not divisible by 4.
2. Thusf is irreducible.
3. Its derivative $5x^4 - 4$ vanishes at $\pm\alpha$ where $\alpha = (4/5)^{1/4}$; it is positive outside and negative inside.
4. Since $\alpha > 5/8$, $f(\alpha) = 2 - (16/5)\alpha < 0$ and $f(-\alpha) = 2 + (16/5)\alpha > 0$.
5. The limits at $-\infty$ and ∞ are $-\infty$ and ∞.
6. Strict monotonicity on the three intervals therefore gives exactly three distinct real roots and two nonreal roots forming one conjugate pair.
7. Irreducibility over Q in characteristic zero supplies separability, so all five complex roots are distinct.

B. §4.N2 — Two permutation types

1. Let G be the splitting-field Galois group acting faithfully on five roots.
2. Irreducibility gives a transitive action, so 5 divides|G| by orbit–stabilizer.
3. Cauchy supplies an element of order 5, necessarily a5-cycle in S_5.
4. Complex conjugation is an automorphism of the splitting field fixing Q; it fixes the three real roots and swaps the nonreal pair, hence is a transposition.

C. §4.N3 — Generate the full symmetric group

1. Label the roots cyclically so σ acts by addition1 modulo 5, and write $\tau = (a, b)$ with nonzero differenceb $-a$ mod 5.
2. Its conjugates $\sigma^j \tau \sigma^{(-j)}$ swap the translated pair $s(a + j, b + j)$.
3. These edges form a connected graph since repeated addition of a nonzero residue visits all five vertices.

4. Edge transpositions on a connected graph generate every transposition: along a path, conjugating the final edge swap by the preceding path swaps gives a swap between the endpoints.

5. All transpositions generateS $_5$, so $S_5 \subset G \subset S_5$ and $G = S_5$.

D. §4.N4 — Nonsolvability and radical obstruction

1. In A_5 the conjugacy-class sizes are 1,15 (double transpositions),20 (3-cycles), and 12,12 (the two classes of 5-cycles).

2. They follow from cycle counting and centralizer sizes4,3,5 in A_5.

3. A normal subgroup is a union of these classes containing the identity, and its order divides60.

4. The possible such sums are 1,13,25,16,28,40,21,33,45,36,48,60; only1 and 60 divide60.

5. Thus A_5 is simple.

6. It is nonabelian, so its nontrivial commutator subgroup, being normal, is all A_5; its derived series never reaches1.

7. Hence A_5, and therefore S_5 which contains it, is not solvable.

8. By the imported radical criterion the quintic is not solvable by radicals.

9. A radical formula for every quintic would solve this one, contradiction.

□

V. WORKED ORCHESTRATION

The critical-point audit uses $\alpha \approx 0.9457$ only as an illustration; the exact inequality $\alpha > 5/8$ proves the signs. By contrastx $^5 - 2$ is irreducible but has roots expressed by a fifth root and roots of unity, so irreducibility alone cannot imply nonsolvability.

A. Boundary and hypothesis audit

A candidate exports "irreducible quintic $\Longrightarrow G = S_5$." Validation counters with $x^5 - 2$ and with smaller transitive groups. The repaired route retains both the 5-cycle and single transposition, proves connected generation, and only then invokes the radical obstruction.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A candidate exports "irreducible quintic $\Longrightarrow G = S_5$." Validation counters with $x^5 - 2$ and with smaller transitive groups. The repaired route retains both the 5-cycle and single transposition, proves connected generation, and only then invokes the radical obstruction.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- MIT Algebra II notes: field extensions and Galois theory.